

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Ionising radiation - Set 1

Mark scheme

AQA spec 3.4.3.2.14; Pollution; textbook pp. 295-304

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• The time taken for half of the atoms of a radioactive isotope to decay (activity halves)	1
01.2	• $32 / 8 = 4$ half-lives • $6400 / 2^4 = 400$ Bq	2
01.3	• Short half-life isotopes (I-131) release all their radiation quickly: dangerous but only briefly so short-term precautions (potassium iodide tablets, evacuation) are sufficient • Very long half-life isotopes (U-238) emit only small amounts of radiation in any period so the level of danger is low • Decades-long half-lives combine a high rate of radiation release with persistence for most or all of a human lifetime, so short-term protection is inadequate (land contaminated for generations)	3
01.4	• Alpha particles have very low penetrating power and are absorbed by clothing / dead skin so external sources cause little exposure • Inside the body the alpha particles are absorbed in a very small mass of tissue so damage is concentrated (high RBE - weighting factor 20) • Long half-life means it continues to emit for the rest of the person's life / cannot be excreted quickly (contamination not just exposure)	3
01.5	• A previously stable nucleus that becomes radioactive after absorbing a neutron (eg in reactor structures), later decaying to emit ionising radiation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 299-301; AQA spec 3.4.3.2.14.	

Question	Marking guidance	Marks
02.1	• Radiation absorbed by tissue produces free radicals (ions with unpaired electrons), mostly from water, which are highly reactive • Free radicals damage DNA / chromosomes in the nucleus causing mutations (cancer, inherited effects) - rapidly dividing cells (skin, gut, bone marrow) most affected • Damage to other cell parts can stop the cell functioning or kill it; low damage may be repaired by the intact nucleus	3
02.2	• Acute: effects appearing quickly after a large dose over a short time (radiation sickness, damage to bone marrow / immune system / gut, haemorrhage, rapid death) • Chronic: effects appearing slowly, in proportion to the dose received over a long period (cancers, cataracts)	2
02.3	• Exposure: absorption of ionising radiation from a source the person is close enough to • Contamination: physically carrying radioactive material on or in the body, so exposure continues wherever the person goes; sealed sources can expose but not contaminate	2
02.4	• Sieverts = grays x radiation weighting factor (20 for alpha) • $0.002 \times 20 = 0.04 \text{ Sv}$ (40 mSv)	2
02.5	• The activity of a source (number of decays per second)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 301, 303; AQA spec 3.4.3.2.14.	
03.1	• Doubling the distance from 1 m to 2 m reduces the dose rate from 400 to 100 (one quarter) • Distance x 10 (1 to 10 m) reduces it by 100 times (400 to 4); dose is proportional to $1/\text{distance}^2$	2
03.2	• $400 / 20^2 = 1 \text{ } \mu\text{Sv h}^{-1}$	1
03.3	• Sealed container prevents contact / contamination so exposure stops when the worker moves away; container may absorb alpha completely; remote handling • Barriers (lead, concrete, thick steel flask walls) absorb radiation before it reaches the worker; thickness depends on radiation type • Dose received is proportional to time near the source so work is organised to minimise time close to it (job rotation) • Dosimeters / film badges record exposure; air monitors detect alpha-emitting particles; contamination monitors on leaving check that no material is carried out	4
03.4	• Stable iodine saturates the thyroid • So radioactive iodine-131 released is not taken up / concentrated in the thyroid gland	2
03.5	• Exposure should be As Low As Reasonably Achievable	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 302-303; AQA spec 3.4.3.2.14.	
04.1	• The underlying granite contains uranium whose decay produces the radioactive gas radon (an alpha emitter) • Radon travels up cracks in the rock and through foundations into buildings • It becomes concentrated inside enclosed / poorly ventilated buildings; control by ventilating foundations / sumps	3

Question	Marking guidance	Marks
04.2	• $16 + 1.5 = 17.5\%$	1
04.3	Any 2 from: • Where they live (geology / radon, altitude) • Occupation (radiographer, nuclear worker, aircrew, miner) • Medical history (number of X-rays / CT scans) • Lifestyle / diet (eg shellfish near discharges); smoking	2
04.4	Any 4 from: • X-rays carry a small risk of mutation / cancer but the benefit (diagnosis, avoiding invasive surgery) may outweigh it; risks can be reduced (shielding, dose limits) • Symptoms of chronic low-dose exposure take a long time to develop • The same symptoms (cancers) have other causes so effects are hard to attribute • Accurate data on human exposure / doses is often unavailable (especially after accidents) • People who benefit and people who take the risk may be different (eg uranium miners vs electricity consumers)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 297-299; AQA spec 3.4.3.2.14.	
05.1	• High-level: used fuel rods / fission products separated in reprocessing; vitrification in molten glass in sealed stainless steel containers, surrounded by concrete, air-cooled to remove decay heat (Sellafield) • Intermediate: fuel cladding / metal tubes around fuel rods, filters from reprocessing; mixed with cement (encapsulated) in stainless steel drums, stored in concrete vaults • Low-level solid: contaminated clothing, equipment; sealed in plastic bags inside steel drums / containers in a concrete-lined, monitored landfill (Drigg)	6
05.2	• Radioactive decay of the concentrated fission products generates heat which could damage containers • Isotopes with long half-lives (plutonium 24 400 years) remain hazardous for thousands of years	2
05.3	• Liquids: filtered and passed through ion-exchange (sands / resins) to remove radionuclides, then discharged (monitored by CPA / CGM) • Gases: filtered (HEPA / charcoal filters) to remove particles and iodine before release	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 303; AQA spec 3.4.3.2.14.	

Question	Marking guidance	Marks
06.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Uses / benefits: nuclear electricity (low carbon, high energy density), medicine (X-rays, CT, radiotherapy, tracers, sterilising equipment), industry (thickness gauges, polymer cross-linking, testing turbines, oil exploration), agriculture (food irradiation, sterile male technique, mutation breeding), research tracers, smoke detectors, ship / submarine propulsion • Risks: free radical production, DNA damage, mutation, cancer; acute and chronic, somatic and gonadic effects; contamination vs exposure; activation products; long-lived wastes; accidents (Chernobyl, Fukushima); weapons fallout • Factors affecting risk: type of radiation (alpha, beta, gamma, neutrons - penetration, RBE / weighting factors), half-life, dose (Gy, Sv), route of exposure • Risk:benefit analysis and why it is difficult (latency, alternative causes, poor exposure data, risks and benefits to different people, offsets such as avoided surgery) • Principles: ALARA, BATNEEC; closed sources, remote handling, absorbers / shielding, distance (inverse square law), time, protective clothing, decontamination, potassium iodide • Monitoring: dosimeters, film badges, air and contamination monitors; environmental monitoring via CPA and CGM; waste categories and disposal (vitrification, encapsulation, Drigg) • Evaluation / judgement: benefits generally outweigh risks when controls are applied; public perception vs actual risk (background radon 50% of dose vs <1% from nuclear industry)	9
06.2	• Consumers of the electricity may live in a different country / region from the uranium miners or the communities near waste stores who bear the risks	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 295-304; AQA spec 3.4.3.2.14.	